Municipal Waste Management



Content

Municipal Waste Management

- Waste Stream and components
- Methods of Waste Disposal
- Recycling and Reusing Waste: Problems & Benefits
- Reducing Waste Stream: Necessity & Urgency
- Hazardous & Toxic Waste
- Sustainable Waste Management
- Circular Economy Concept

Waste and the Waste Stream

Garbology: the science of trash

What can the stuff we throw away every day tell us about our society – and ourselves?
Professor William Rathje, UA

- Consumption pattern
- perception on waste from cradle –to-grave
- Behavior



Waste and the Waste Stream

Why do we waste things?

- It's no more useful
- It's unwanted
- It's defective
- It's old & worthless

Waste: unwanted by-products and residues in nearly everything we do.

Waste Stream: is the term that describes the steady flow of varied wastes that we all produce, from domestic garbage and yard wastes to industrial, commercial, and construction refuse.

Waste and the Waste Stream

Waste streams can be divided into two broad types:

✓ Material-related streams (such as metals or plastics)
✓ Product-related streams (such as electronic waste or end-of-life vehicles)



eurostat

Waste and the Waste Stream

Sources of Waste – Global Classification

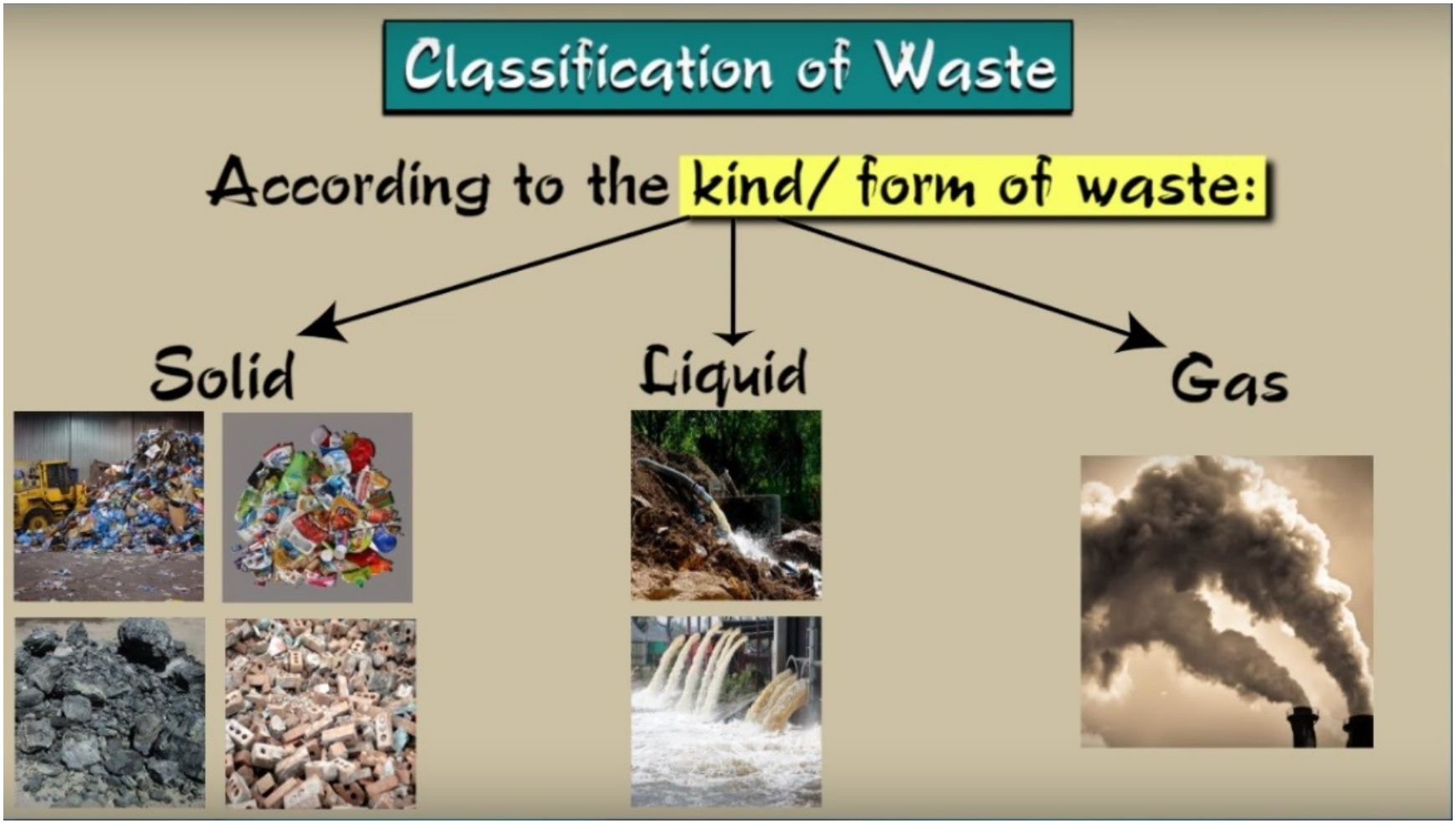
- Household – MSW domestic
- Public spaces – MSW public – public bins, legal dumping activities etc.
- Business – C&I – waste generated by business and industrial productions
- Construction – construction and demolition residual
- Disaster – waste generated from unplanned events (e.g., flood, cyclone, earthquake, fire hazard etc.) Mandalay

Waste Management



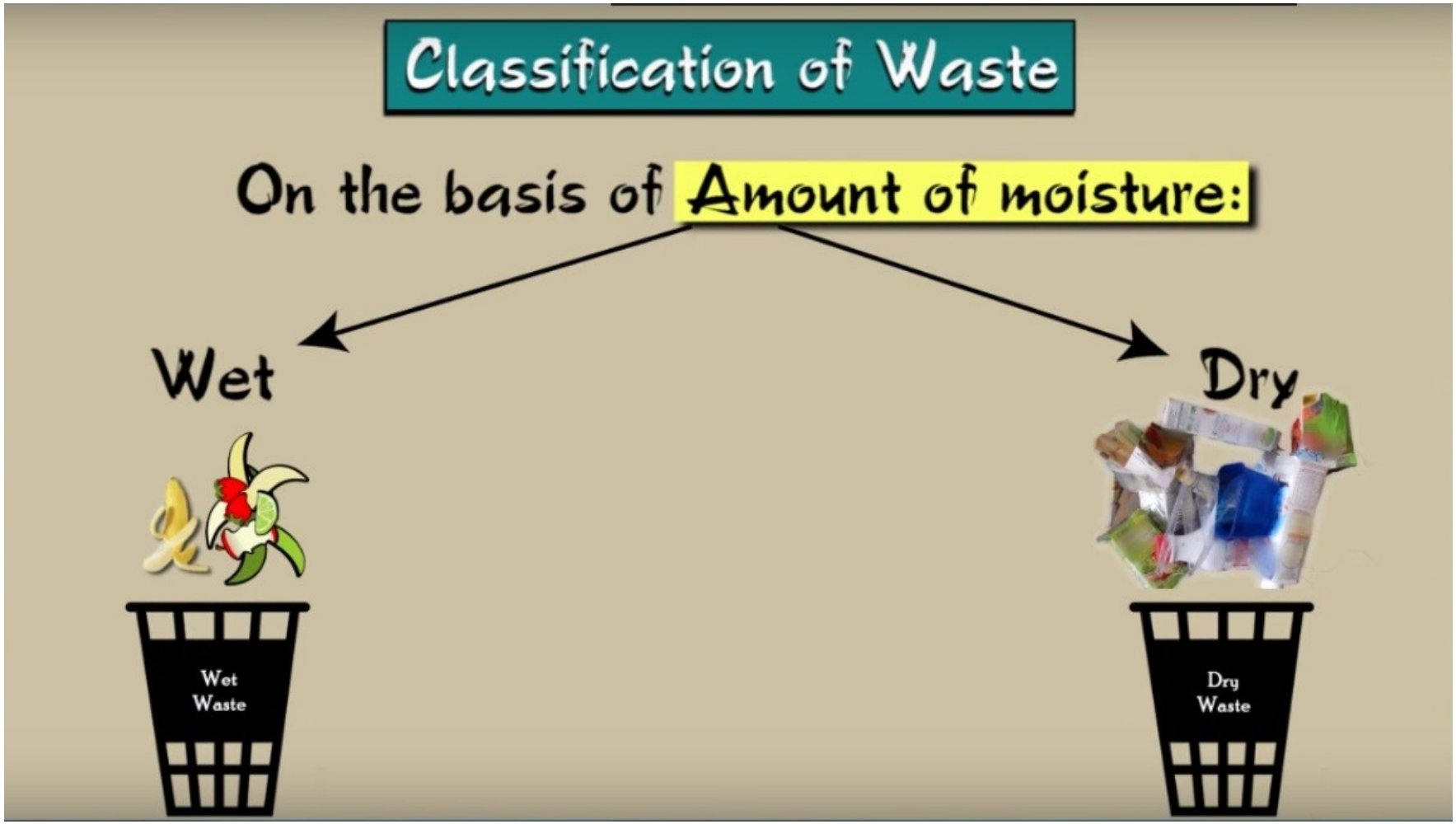
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Waste and the Waste Stream



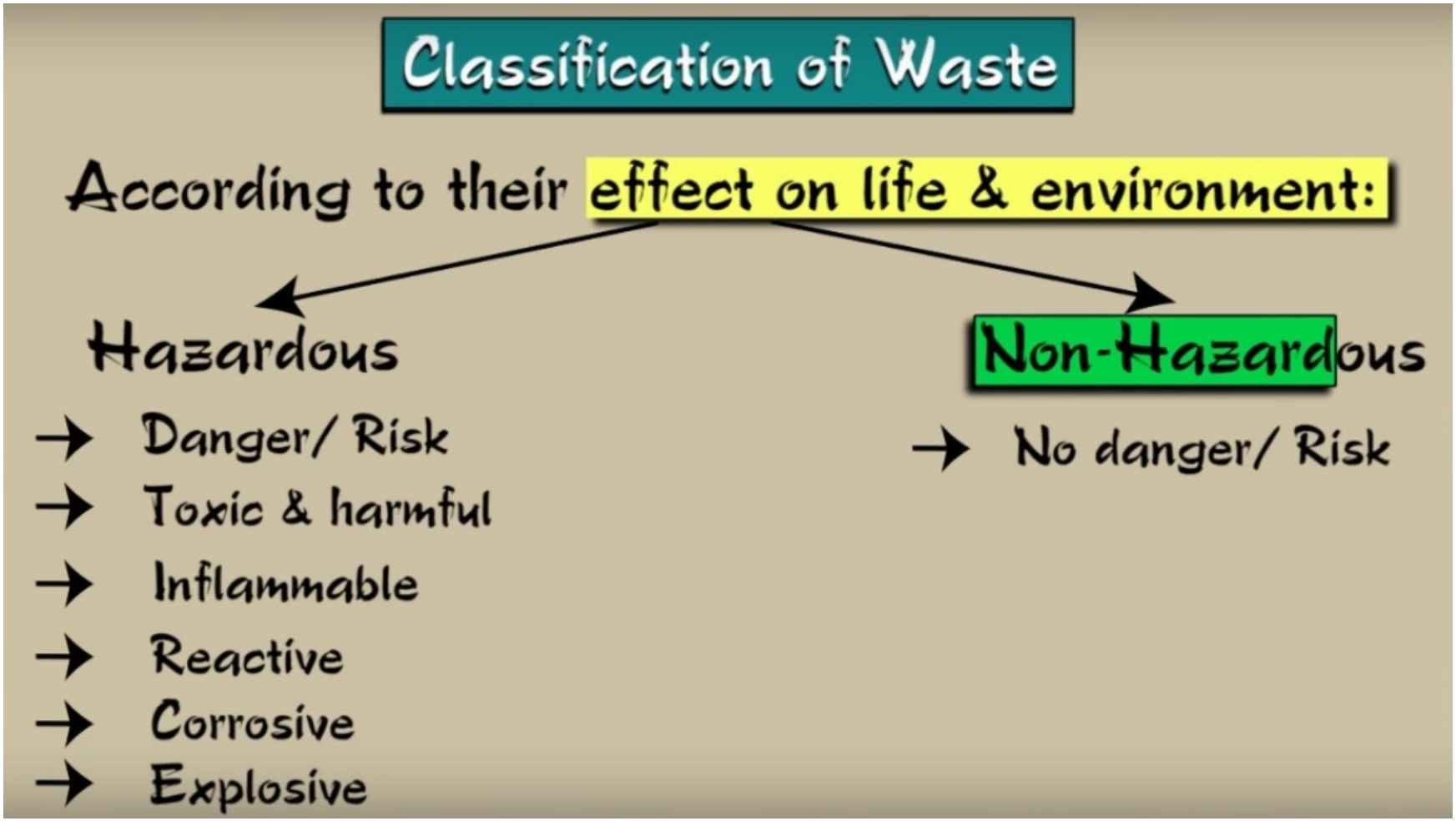
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Waste and the Waste Stream



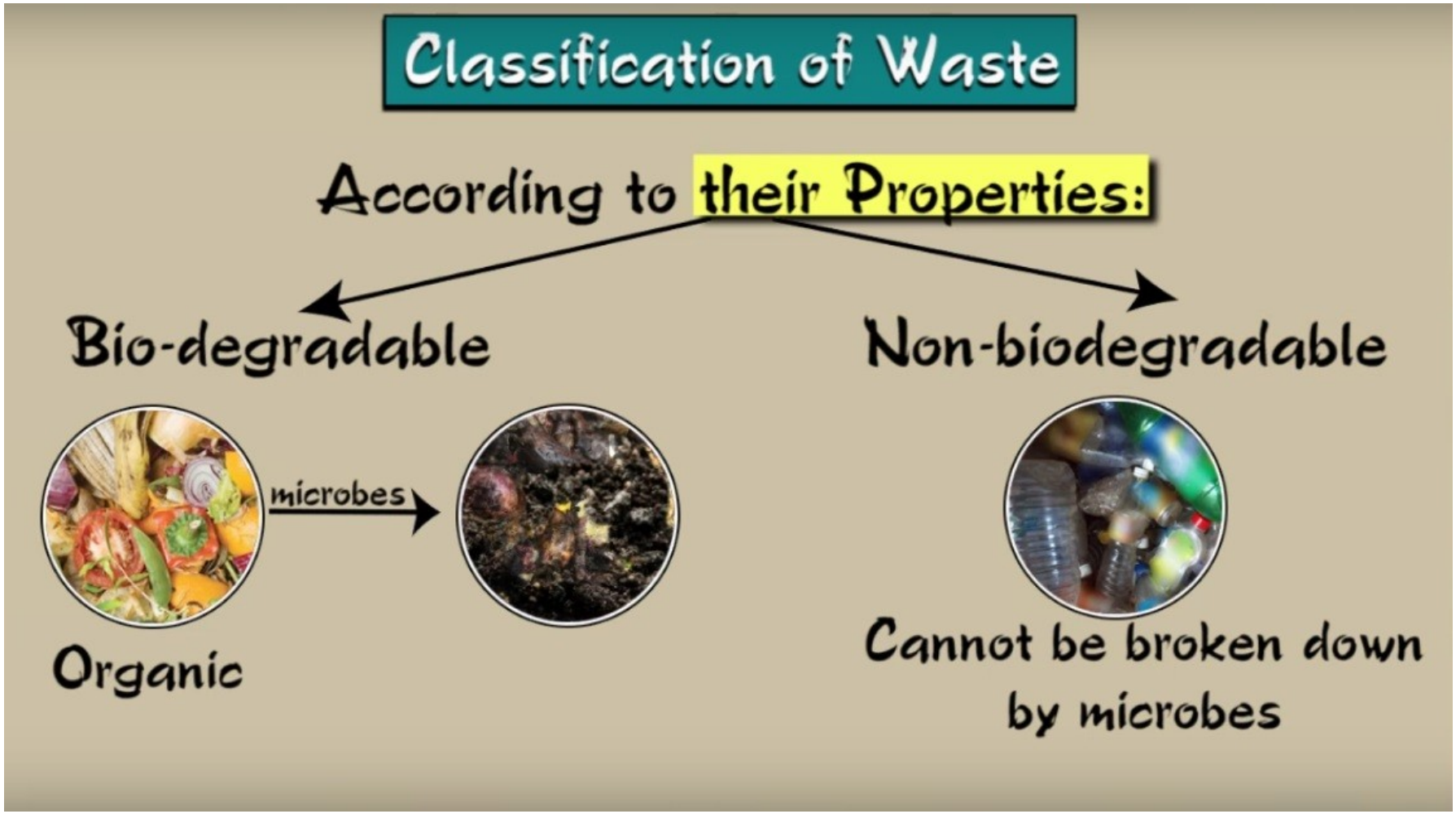
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Waste and the Waste Stream



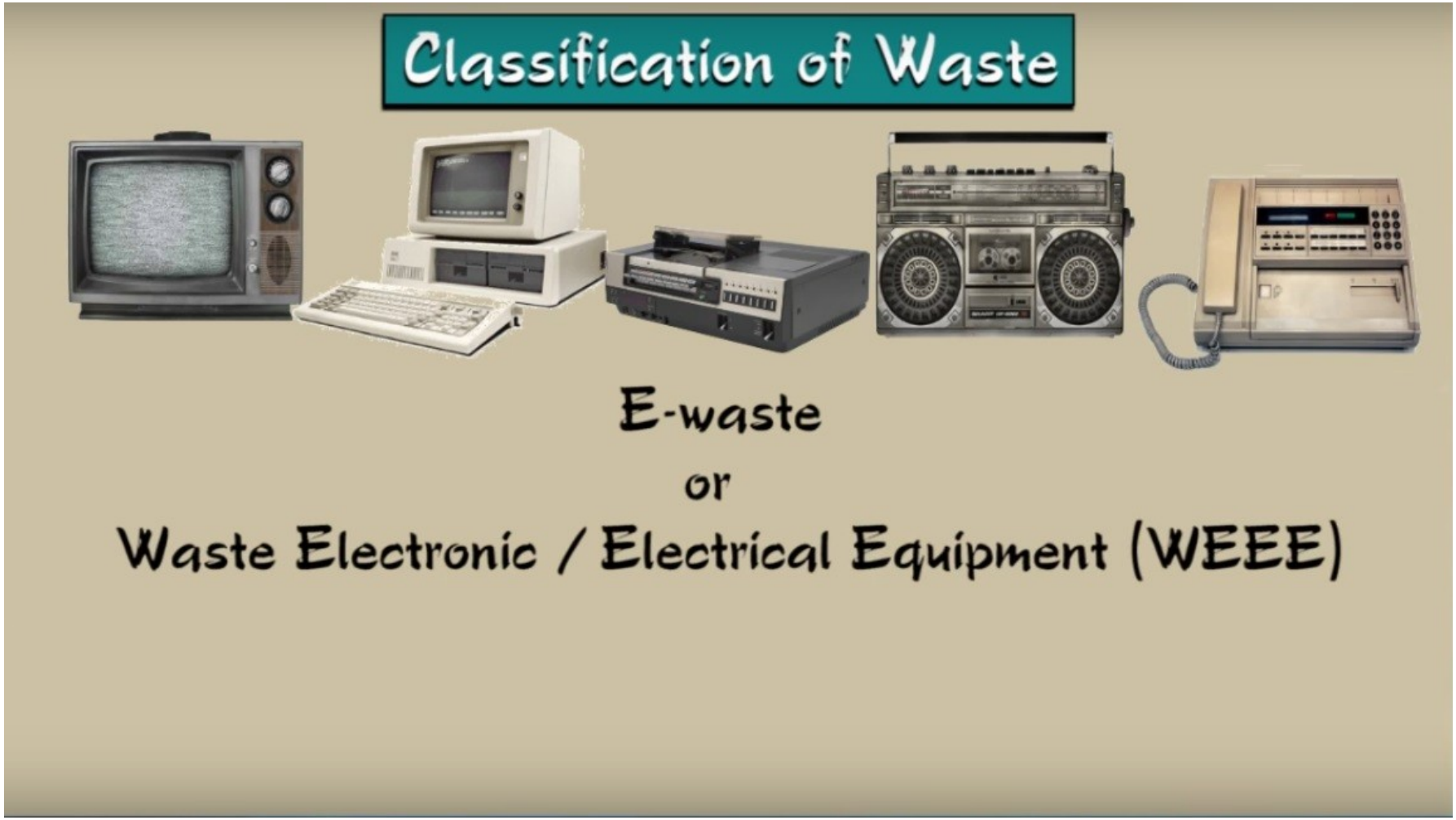
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Waste and the Waste Stream



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Waste and the Waste Stream

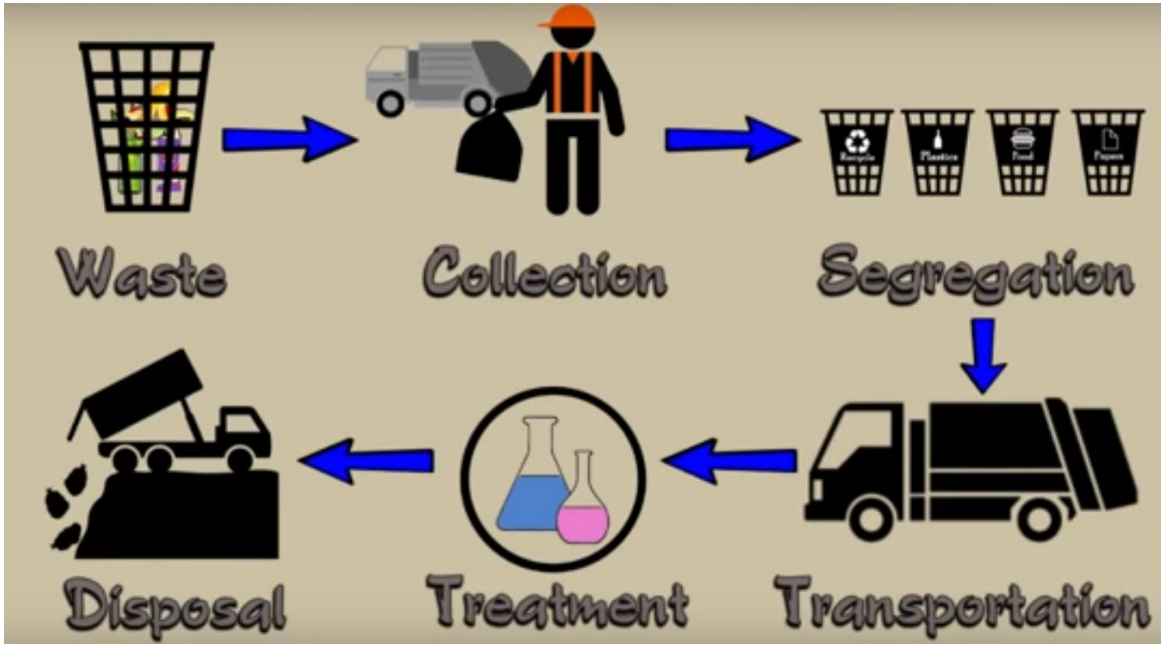


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Municipal Waste Management

Municipal waste: (also referred to as 'municipal solid waste') waste from households, as well as other waste which, because of its nature or composition, is similar to waste from households. (EPRS)

Municipal Solid Waste – Residential and commercial solid waste generated within a community. (UNEP)



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Waste Management

Need to segregate



Need to transport

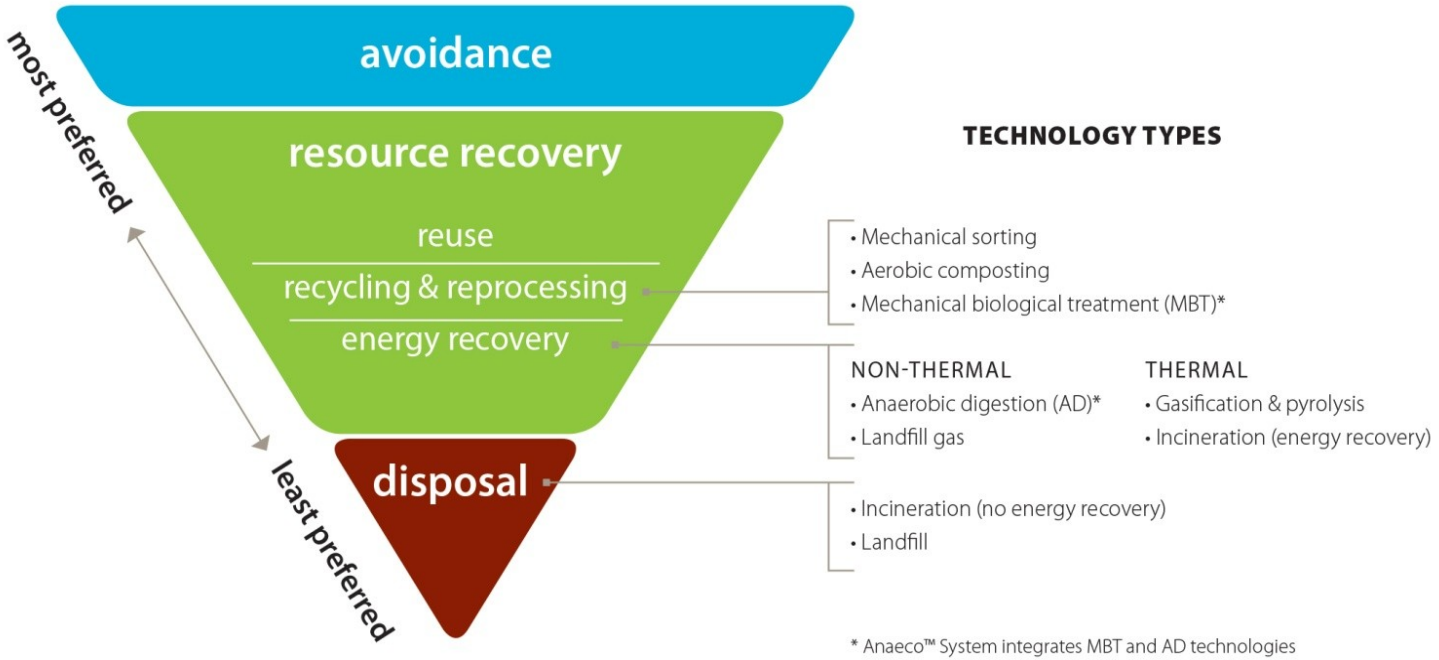


Waste Management

Two principles apply across all waste streams:

The **'waste hierarchy'** sets a priority order among five waste treatment methods:
1) prevention, then 2) preparing for re-use,
3) recycling and 4) (energy) recovery, with
5) disposal as the least preferred option.

European Model



Waste Management

The **'polluter pays principle'** requires polluters to bear the cost of preventing, controlling and cleaning up pollution, to ensure that these costs are reflected in the price of goods and services causing pollution at the production and/or consumption stage.

Implementing these principles, **extended producer responsibility (EPR)** is used in several waste Streams. It implies that producers take over the (financial and/or
operational) responsibility for collecting or taking back used goods, for sorting them and for their eventual recycling.

Three 'R's:

Modern waste management method follows the three 'R's which means

- **Reduce -** the amount and toxicity of trash that discard.
- **Reuse -** containers and products; repairing what is broken or giving it to someone who can repair it.
- **Recycle -** as much as possible, which means buying products with recycled content.

Waste Management

think about the **benefits** and challenges of **3R**



Waste Management

Disposal Methods:

Open dump – drop it some place **Not desirable**

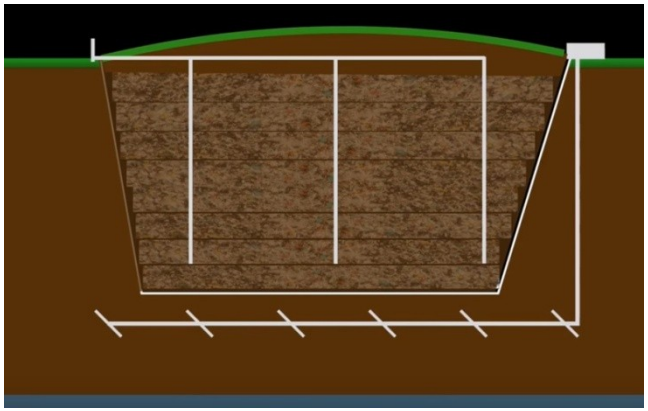
Land fill – dig, fill, separate, compact, secure & cover

Most used

Ocean dump – dump in ocean **Not desirable**

Incineration – burn & recover **Used**







Waste Management

Waste Scoreboard - **EUROPEAN COMMISSION, 2012**

Criterion / EU MS	1.1 Decoupling of waste from consumption	1.2 Waste Prevention Programme	1.3 Amount of municipal waste recycled	1.4 Amount of municipal waste recovered (energy recovery)	1.5 Amount of municipal waste disposed	1.6 Development of municipal waste recycling	2.1 Existence of ban/restrictions for the disposal of municipal waste into landfills	2.2 Total typical charge for the disposal of municipal waste in a landfill	2.3 Existence of pay-as-you-throw (PAYT) systems for municipal waste	3.1 Access to waste collection services	3.2 Available treatment capacity for municipal waste	3.3 Forecast of municipal waste generation and treatment capacity in the Waste Management Plan	3.4 Existence and quality of projection of municipal waste generation and treatment	3.5 Compliance of existing landfills for non-hazardous waste	4.1 Fulfilment of the targets related to biodegradable municipal waste going to landfills	4.2 Rate of biodegradable municipal waste going to landfills	5.1 Number of infringement procedures – WFD and Landfill Directives	5.2 Number of court cases – WFD and Landfill Directives	Over all score
AT	0	2	2 D	2 D	2 D	2	2	1	2	2	2	2	2	2	2	2	2	2	39
NL	0	2	2 D	2 D	2 D	2	2	2	1	2	2	2	2	2	2	2	2	2	39
DK	0	0	2 D	2 D	2 D	2	2	2	1	2	2	2	2	2	2	2	2	2	37
DE	1	0	2 D	1 D	2 D	2	2	2	2	2	2	2	1	2	2	2	2	2	36
SE	1	2	2 D	2 D	2 D	2	2	2	1	2	2	0	0	1	2	2	2	2	35
BE	1	2	2 D	2 D	2 D	2	2	2	1	2	2	0	0	2	2	2	1	1	34
LU	0	0	2 D	2 D	2 D	2	2	2	1	2	2	0	0	2	2	2	2	2	33
UK	1	2	2 D	1 D	2 D	2	0	1	1	2	2	2	1	1	2	1	2	2	32
FI	1	2	1 D	2 D	1 D	0	1	1	2	2	2	2	1	1	2	2	2	2	31
FR	1	2	1 D	2 D	2 D	1	1	1	1	2	2	2	1	1	2	2	1	1	31
SI	2	0	2 D	1 D	1 D	2	1	2	2	0	2	0	0	0	2	1	1	2	25
ES	2	0	1 D	1 D	1 D	1	0	1	1	2	2	0	0	1	2	1	1	1	21
PT	0	2	0 D	2 D	1 D	1	0	0	0	2	2	2	2	2	0	0	1	1	21
HU	1	0	1 D	1 D	1 D	2	0	0	1	0	0	0	0	2	2	1	2	2	19
IE	0	2	1 D	1 D	1 D	1	1	2	1	0	2	2	0	2	0	0	0	0	19
CZ	2	0	0 D	1 D	1 D	2	0	1	1	2	2	0	0	1	0	0	1	2	18
PL	1	2	1 D	0 D	1 D	2	1	1	1	0	2	0	0	1	0	0	1	2	18
EE	2	0	1 D	0 D	0 D	0	1	1	1	0	2	0	1	2	2	1	1	1	17
SK	2	0	0 D	1 D	0 D	1	1	0	1	2	2	0	0	1	2	1	1	1	17
IT	0	0	1 D	1 D	1 D	0	1	2	1	2	0	0	0	0	2	1	0	0	15
LV	2	0	0 D	0 D	0 D	1	1	1	0	0	2	0	1	2	0	0	2	2	14
CY	0	0	1 D	0 D	0 D	2	0	0	0	2	0	0	1	0	0	0	2	2	11
RO	2	0	0 D	0 D	0 D	1	0	0	0	0	2	0	1	0	0	1	2	2	11
LT	2	0	0 D	0 D	0 D	1	1	0	1	0	0	0	0	0	0	0	2	2	9
MT	0	0	0 D	0 D	0 D	2	0	0	0	2	0	0	0	2	0	0	1	2	9
BG	2	0	0 D	0 D	0 D	0	0	0	0	0	0	0	0	0	2	1	1	2	8
GR	1	0	0 D	0 D	0 D	0	0	0	0	2	0	0	0	0	0	0	0	0	3

Waste Management

Waste Scoreboard - **EUROPEAN COMMISSION, 2012**

Criteria in full

1.1 Level of decoupling of municipal waste generation from household final consumption expenditure
1.2 Existence of own waste prevention program (WPP) or equivalent existence in Waste Management Plan or other (environmental) program
1.3. Amount of municipal waste recycled (material recycling and other forms of recycling including composting)
1.4. Amount of municipal waste recovered (energy recovery)
1.5. Amount of municipal waste disposed (deposit onto or into land and incinerated without energy recovery)
1.6. Development of municipal waste recycling (material recycling and other forms of recycling including composting)
2.1. Existence of nationwide ban/restrictions for the disposal of municipal waste into landfills
2.2. Total typical charge for the disposal of municipal waste in a landfill
2.3. Existence of pay-as-you-throw (PAYT) systems for municipal waste
3.1. Access to waste collection services for municipal waste
3.2. Available treatment capacity for municipal waste in line with the EU waste legislation (including disposal and incineration)
3.3. Forecast of municipal waste generation and treatment capacity in the Waste Management Plan
3.4. Existence and quality of projection of municipal waste generation and treatment in the Waste Management Plan
3.5. Compliance of landfills for non-hazardous waste with Landfill Directive
4.1 Fulfilment of the targets of the Landfill Directive related to biodegradable municipal waste going to landfill
4.2. Rate of biodegradable municipal waste going to landfills
5.1. Number of infringement procedures – Waste Framework Directive and Landfill Directive
5.2. Number of court cases – Waste Framework Directive and Landfill Directive

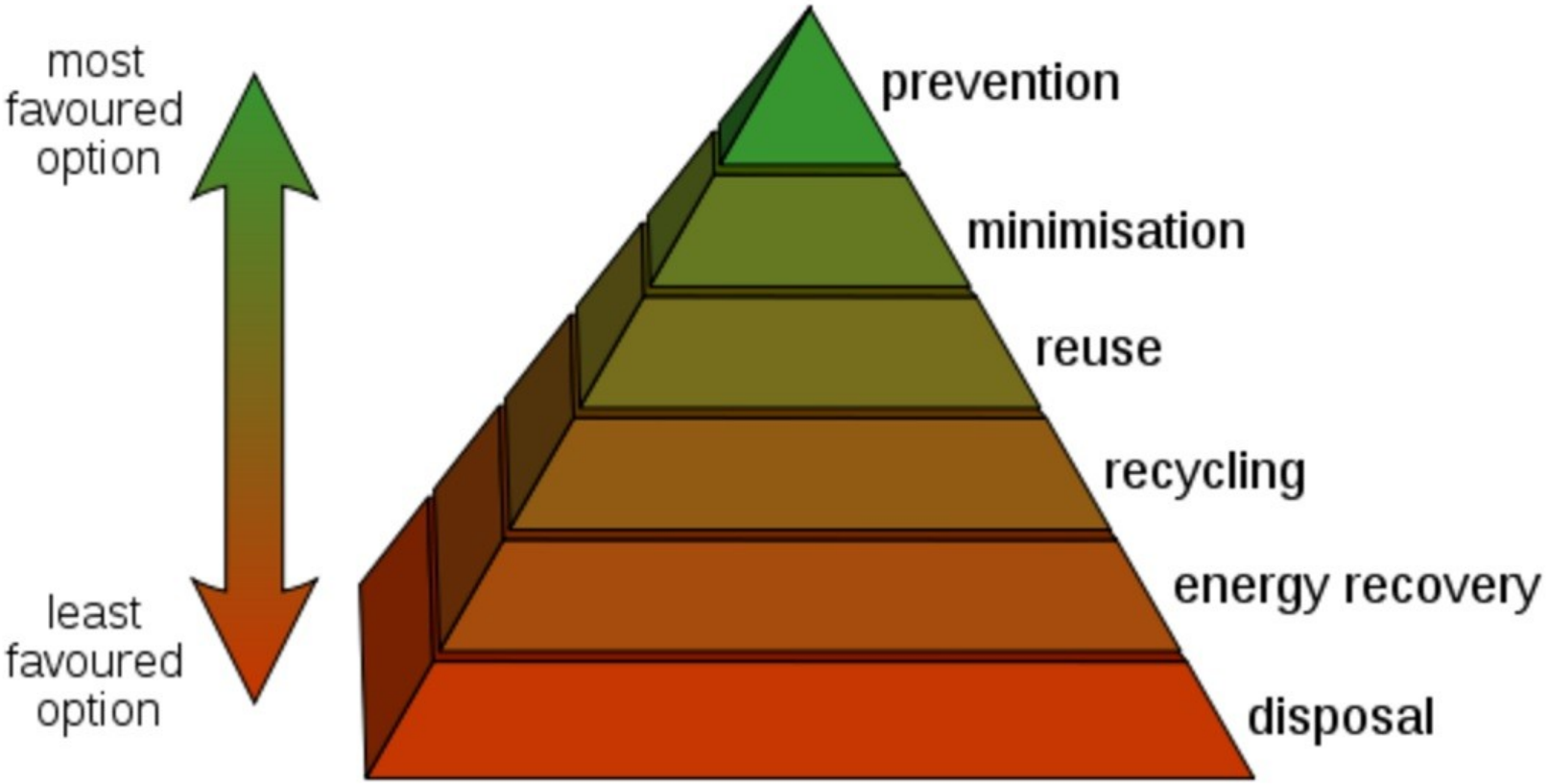


Sustainable Waste Management

Sustainability means taking into account in our work and in our daily lives all the resources upon the planet: humans, animals, plants, water, air, land space [etc] and managing them to grow and develop the resources that today exist, for the future.

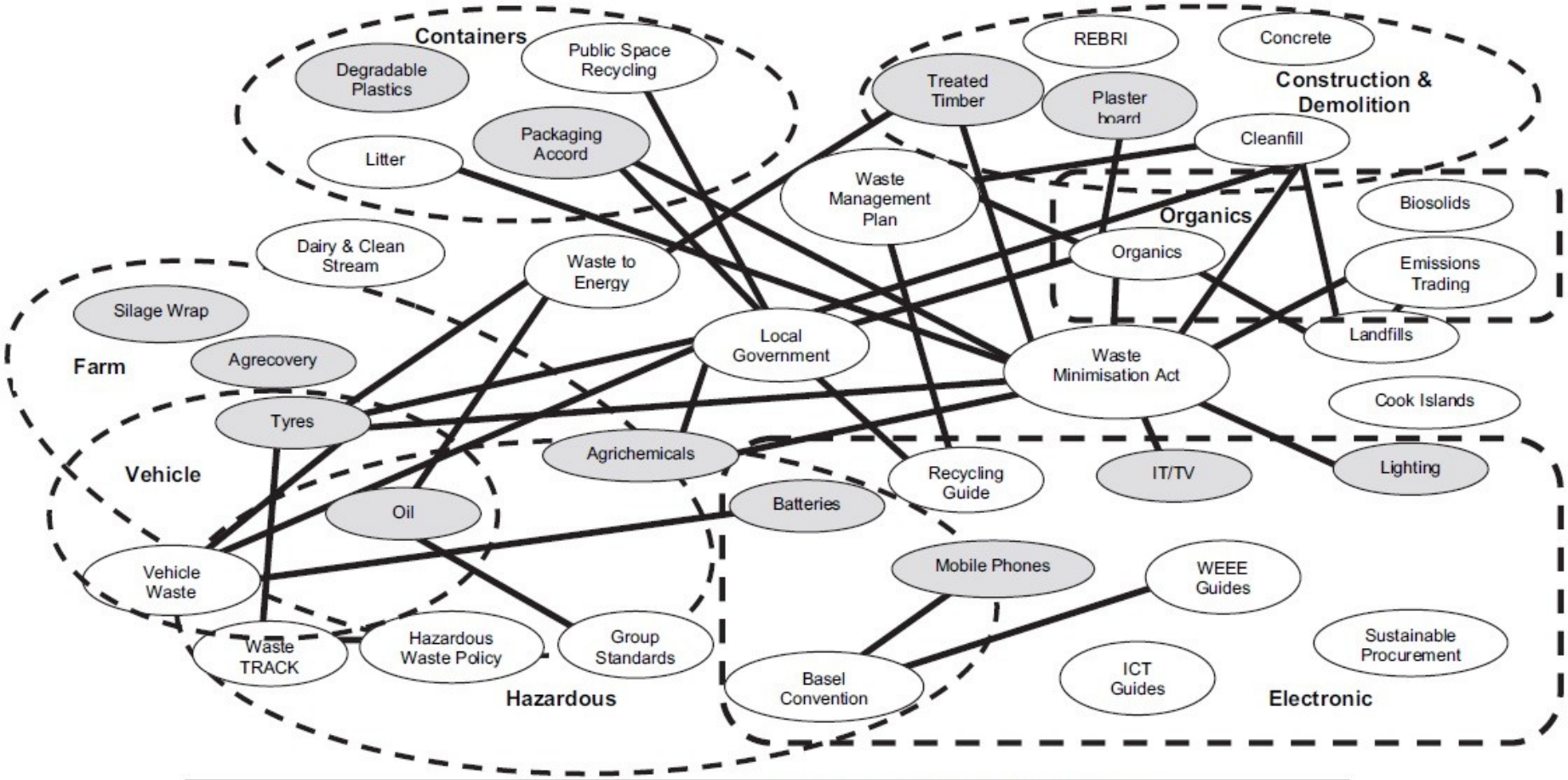
Using material resources efficiently to cut down on the amount of waste produced. And, where waste is generated, dealing with it in a way that actively contributes to the economic, social and environmental goals of sustainable development.

Sustainable Waste Management



MONICA PIANOSI

SWM - Complexity of Waste Management



J.K. Seadon, 2010

Sustainable Waste Management

Approaches to look at the issue…**many**

Sustainable Approach by GIZ

- Waste-to-energy technologies
- Avoidance of marine litter
- Electronic waste
- Economic instruments

giz Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) GmbH



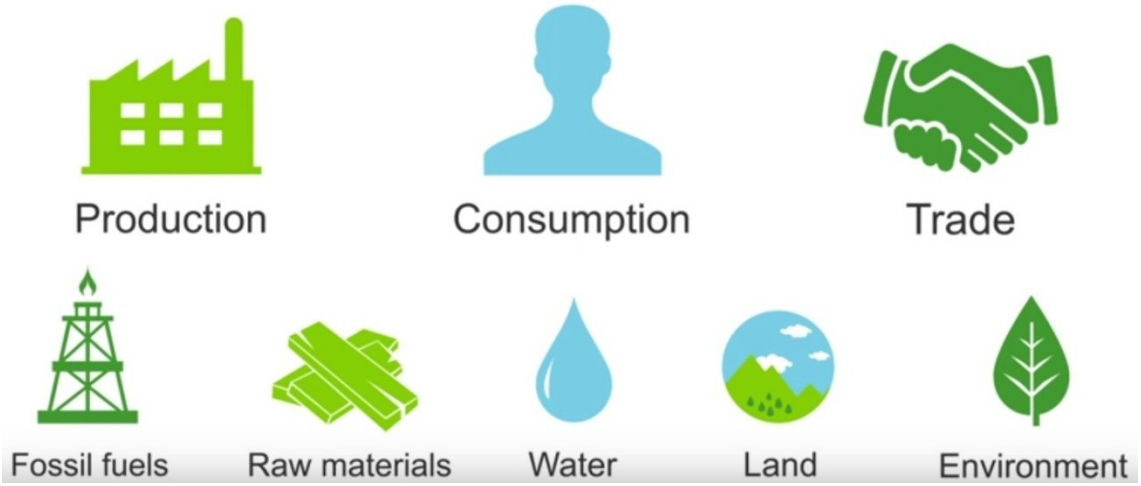
Sustainable Waste Management

Instead of Approaches….

…it could be an **economic theory!!!**

Sustainable Waste Management

"**Circular Economy** – can be defined as a system in which production, consumption and trade are designed to minimize the net extraction of key resources; such as fossil fuel, raw materials, water, land and the environment in pollutants along with diminishing negative environmental and health impacts" EU





Sustainable Waste Management

Today's linear 'take, make, dispose' economic model relies on large quantities of cheap, easily accessible materials and energy, and is a model that is reaching its physical limits.

A **circular economy** is restorative and regenerative by design, and aims to keep products, components, and materials at their highest utility and value at all times.

Principle (s)

Principle 1: Preserve and enhance natural capital

...by controlling finite stocks and balancing renewable resource flows.

Principle 2: Optimize resource yields

...by circulating products, components, and materials at the highest utility at all times in both technical and biological cycles.

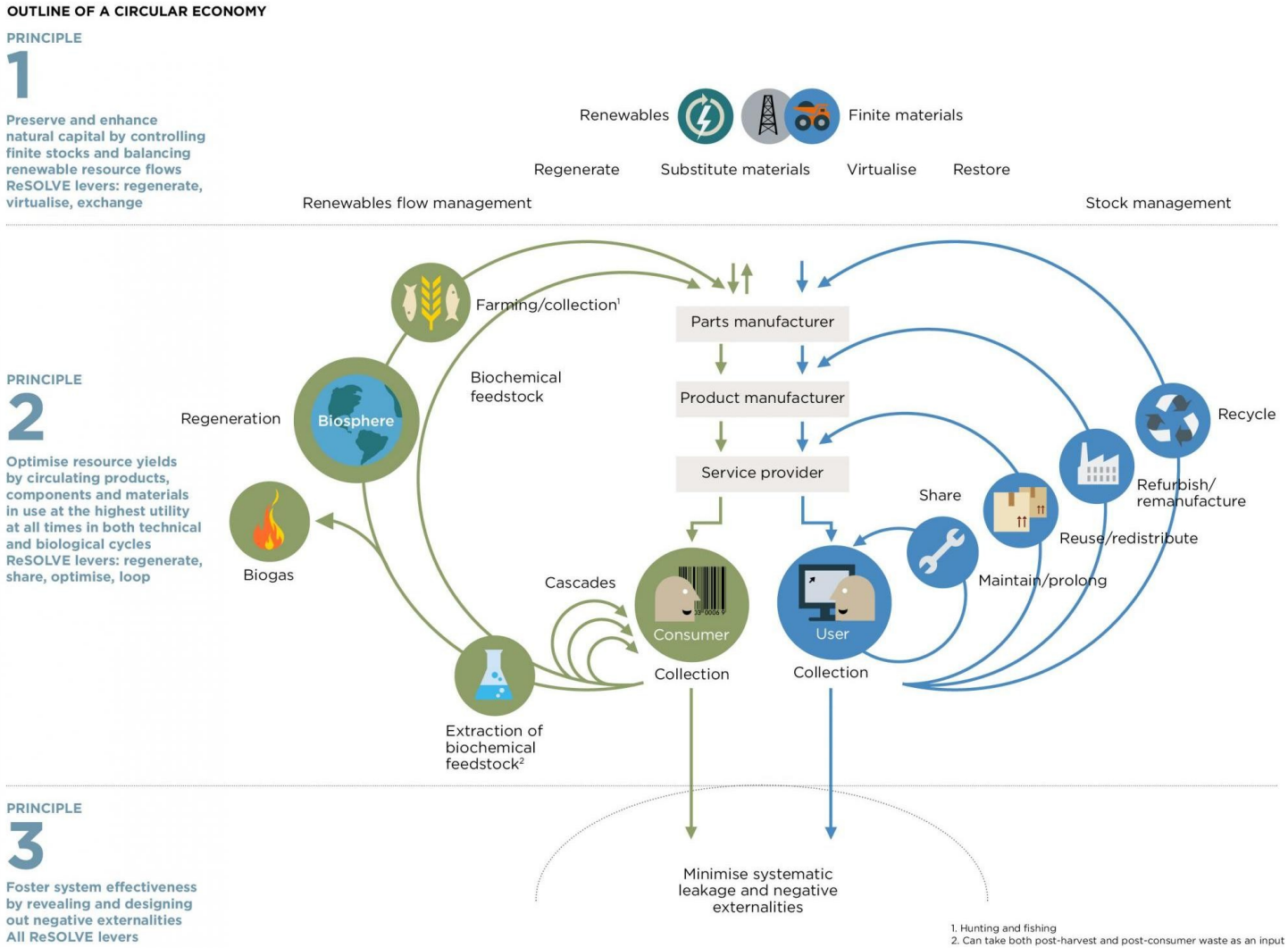
Principle 3: Foster system effectiveness

...by revealing and designing out negative externalities.

Ellen Macarthur Foundation
https://www.ellenmacarthurfoundation.org/circular-economy

Sustainable Waste Management

Circular Economy Model



Ellen Macarthur Foundation
https://www.ellenmacarthurfoundation.org/circular-economy

Thanks

